PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Male partner unknown HIV status as a risk factor for HIV incidence and clinical outcomes in Prevention of Mother-to-Child Transmission of HIV programmes in 21 WHO priority countries, a systematic review protocol
AUTHORS	Mmotsa, Tshiamo; Ngandu, Nobubelo; Adetokunboh, Olatunji; Nyasulu, P

VERSION 1 – REVIEW

REVIEWER	Julia Dettinger University of Washington , Department of Global Health
REVIEW RETURNED	06-Oct-2021

GENERAL COMMENTS	This is an extremely important topic area and I am so glad to see someone is planning a systematic review and meta-analysis looking at the impact of male partner HIV status knowledge on maternal and child outcomes. My primary comments are: 1. The introduction is so long. It takes 5 detailed, heavily researched paragraphs, to even begin talking about the role of male partner status knowledge on maternal HIV outcomes. I would strongly recommend cutting the introduction significantly, the important of this topic is lost in the lengthy introduction. 2. Many of the citations and initiatives referenced are from the early 2010's and it makes the introduction feel a bit stale. Considering going back and trying to bring in more recent references throughout the document. Minor questions/comments: 1. I am curious why you are not also including clinicaltrials.gov or CROI conference abstracts in your data collection?

REVIEWER	Simukai Shamu APHRC
REVIEW RETURNED	2-Dec-2021

GENERAL COMMENTS	Overall comments This is a well written article about the systematic review protocol that seeks to determine whether not knowing male partner's HIV status is a risk factor for HIV incidence and poor clinical outcomes in pregnant women in WHO high burden countries of Africa. The article is researching a critical aspect at a time when countries battle to meet the 95-95-95 UN goals, which is critical for men and women to work together through disclosure for better outcomes. The article also proposes the review and meta analysis through a rigorous set

	of methods based on best practices and models as well as clearly outlined steps involving the research team. Below are a few areas that need attention to perfect the article. Introduction 1. Putting subheadings in the introduction can help to follow the long introduction. 2. You can also reference fear of disclosure as a risk factor on page 3. Methods 3. In the exclusion criteria section, where you said qualitative interviews should have been qualitative studies… 4. It would be good to provide more information on why the 22 countries are regarded as priority countries and the table showing these countries can be cited. 5. The methods section does not seem to cite any references while a lot of concepts are borrowed from elsewhere, except only on the quality of the manuscripts section where citations are provided 6. Check if the levels of heterogeneity (%) are not overlapping 7. I expected a section on discussion. It might be that the journal does not require this section but in practice, this section would be needed to offer some discussion of the proposed methods and also highlighting possible challenges and how these challenges would be dealt with, etc.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Julia Dettinger, University of Washington
Comments to the Author:
Overview: This is an extremely important topic area and I am so glad to see someone is planning a systematic review and meta-analysis looking at the impact of male partner HIV status knowledge on maternal and child outcomes. My primary comments are:

Comment - The introduction is so long. It takes 5 detailed, heavily researched paragraphs, to even begin talking about the role of male partner status knowledge on maternal HIV outcomes. I would strongly recommend cutting the introduction significantly, the important of this topic is lost in the lengthy introduction.
Response: The introduction has been shortened as recommended from 1411 words to 949 words, to bring the focus only to the burden of HIV in the population of interest, priority countries for eliminating HIV and the risk factor of interest and how it links to other related risk factors. Sub-headings have also been included to make reading easier (see lines 100, 125, 150). We have also made changes to the reference list because of the edits made to the introduction section.

Comment - Many of the citations and initiatives referenced are from the early 2010's and it makes the introduction feel a bit stale. Considering going back and trying to bring in more recent references throughout the document.
Response: More recent references have been added as follows:
Reference 5: UNAIDS. Start Free, Stay Free, AIDS Free. Final report on 2020 targets: July 2021. https://www.aidsdatahub.org/resource/start-free-stay-free-aids-free-final-report-2020-targets date accessed 06 January 2021
Reference 15: van Lettow M, Cataldo F, Landes M, et al. Impact of inter-partner HIV disclosure patterns in Malawi's PMTCT program: A mixed-method study. PloS One 2019;14(7):e0219967.

Reference 23: Trafford Z, Gomba Y, Colvin CJ, Iyun VO, Phillips TK, Brittain K, Myer L, Abrams EJ, Zerbe A. Experiences of HIV-positive postpartum women and health workers involved with community-based antiretroviral therapy adherence clubs in Cape Town, South Africa. BMC Public Health 2018 Dec;18(1):1-3.

Previous reference 7, 26 & 29 have been removed along with references that were part of the text which has been removed from the introduction

Minor questions/comments:
Comment - I am curious why you are not also including clinicaltrials.gov or CROI conference abstracts in your data collection?
Response: Thank you for the recommendation. We have included these two databases to the list of Information sources – see line 204 - 207.

Reviewer: 2
Dr. Simukai Shamu, APHRC
Comments to the Author:
Overall comments: This is a well written article about the systematic review protocol that seeks to determine whether not knowing male partner's HIV status is a risk factor for HIV incidence and poor clinical outcomes in pregnant women in WHO high burden countries of Africa. The article is researching a critical aspect at a time when countries battle to meet the 95-95-95 UN goals, which is critical for men and women to work together through disclosure for better outcomes. The article also proposes the review and meta-analysis through a rigorous set of methods based on best practices and models as well as clearly outlined steps involving the research team. Below are a few areas that need attention to perfect the article.
Response: Thank you for your useful comments.

Comment - Introduction: Putting subheadings in the introduction can help to follow the long introduction.
Response The introduction has been shortened as recommended from 1411 words to 949 words, to bring the focus only to the burden of HIV in the population of interest, priority countries for eliminating HIV and the risk factor of interest and how it links to other related risk factors. Sub-headings have also been included to make reading easier (see lines 100, 125, 150). We have also made changes to the reference list because of the edits made to the introduction section.
Comment - You can also reference fear of disclosure as a risk factor on page 3.
Response: Reference for fear of disclosure has been added, van Lettow et al. [15], see line123.
Reference 15: van Lettow M, Cataldo F, Landes M, et al. Impact of inter-partner HIV disclosure patterns in Malawi's PMTCT program: A mixed-method study. PloS One 2019;14(7):e0219967.

Comment - Methods:
In the exclusion criteria section, where you said qualitative interviews should have been qualitative studies
Response: This has been corrected, we have now written qualitative studies. See line 193.

Comment - It would be good to provide more information on why the 22 countries are regarded as priority countries and the table showing these countries can be cited.
Response: Thanks for this comment. We have included an explanation describing the reasons behind why these 22 countries are regarded as a priority. "These countries also accounted for 90% of the world's HIV positive cases in pregnant women and in a 2020 report most of these countries accounted for more than two thirds up to one fifth of the newly infected children.", see lines 108 to 110 with a more recent reference added - Reference 5 - UNAIDS. Start Free, Stay Free, AIDS Free. Final

report on 2020 targets: July 2021. https://www.aidsdatahub.org/resource/start-free-stay-free-aids-free-final-report-2020-targets date accessed 06 January 2021.
The list of countries in Table 1 has been cited accordingly with reference 4, see line 196. Reference 4: World Health Organisation. Use of Antiretroviral Drugs for treating Pregnant Women and preventing HIV infections in Infants. 2012. https://www.who.int/hiv/PMTCT_update.pdf date accessed 26 January 2020

Comment - The methods section does not seem to cite any references while a lot of concepts are borrowed from elsewhere, except only on the quality of the manuscripts section where citations are provided
Response: Thanks for the observation. We have now appropriately included relevant citation to our methods description. These citations are listed below:
Reference 4: World Health Organisation. Use of Antiretroviral Drugs for treating Pregnant Women and preventing HIV infections in Infants. 2012. https://www.who.int/hiv/PMTCT_update.pdf date accessed 26 January 2020
Reference 27: Moher D, Shamseer L, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. Syst Rev2015;4(1):1-9.

Reference 28: Jackson R, Ameratunga S, Broad J, Connor J, Lethaby A, Robb G, et al. The GATE frame: critical appraisal with pictures. Evidence-based medicine. 2006;11(2):35-8.

Reference 29: Ouzzani M, Hammady H, Fedorowicz Z, Elmagarmid A. Rayyan-a web and mobile app for systematic reviews. Syst Rev. 2016. org

Reference 30: Higgins JP, Altman DG, Gotzsche PC, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. BMJ 2011;343:1–9. https:// doi.org/10.1210/jcem-43-6-1203.

Reference 31: Sterne JAC, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomized studies of interventions. BMJ 2016;355;i4919; doi: 10.1136/bmj.i4919.

Reference 32: Hayden JA, Coté P, Bombardier C. Evaluation of the Quality of Prognosis Studies in Systematic Reviews. Ann Intern Med 2006;144(6):427-438
Reference 33: Higgins JP, Thompson SG, Deeks JJet al. Measuring inconsistency in meta-analyses. BMJ 2003;327(7414):557.

Reference 34: StataCorp. 2019. Stata Statistical Software: Release 16. College Station, TX: StataCorp LLC.

Reference 35: Akers J, Aguiar-Ibáñez R, Baba-Akbari A. Systematic reviews: CRD's guidance for undertaking reviews in health care.

Reference 36: Iorio A, Spencer FA, Falavigna M, et al. Use of GRADE for assessment of evidence about prognosis: rating confidence in estimates of event rates in broad categories of patients. BMJ 2015;350.

Reference 37: Foroutan F, Guyatt G, Zuk V, et al. GRADE Guidelines 28: Use of GRADE for the assessment of evidence about prognostic factors: rating certainty in identification of groups of patients with different absolute risks. J Clinl Epidemiol 2020;121:62-70.

Comment - Check if the levels of heterogeneity (%) are not overlapping

Response: The thresholds for heterogeneity have been edited. This statement now reads “We will consider I^2 statistical values between 30% to 60% as moderate heterogeneity; >60% to 90% as substantial heterogeneity; and >90 to 100% as considerable heterogeneity”, see line 270 to 272.

Comment - I expected a section on discussion. It might be that the journal does not require this section but in practice, this section would be needed to offer some discussion of the proposed methods and also highlighting possible challenges and how these challenges would be dealt with, etc.
Response: Thanks for the suggestion. The discussion section has been added. It reads “This systematic review will provide an overview of how lack of disclosure of male partner’s HIV status to pregnant and postpartum women affects progress towards eliminating HIV incidence and reducing HIV morbidity in pregnant and postpartum women and their infants. This systematic review will arguably be the first to explore the risk of male partner unknown HIV status on HIV incidence and clinical outcomes in PMTCT programmes in the 21 WHO priority countries. As a result of a clearly outlined protocol, there will be minimum duplication of data. This protocol also ensures that a clear and transparent procedure is used. Some limitations are foreseen in this study, such as the non-inclusion of qualitative studies in the meta-analysis methodology to be used, potentially low methodological quality of studies from grey literature (e.g., policy reports) and low statistical power for meta-analysis where few included studies report on the same outcome. However, the study will still help to highlight knowledge gaps in the WHO priority countries about the role of knowing a male partner’s HIV status in improving PMTCT program effectiveness. There is generally scarcity in randomized and non-randomized trials assessing the impact of male involvement in MTCT of HIV programmes as well as studies specifically focused on the impact of lack of knowledge of male partner HIV status on maternal as well as child outcomes. This systematic review will therefore likely pave the way for more primary studies to explore male partner HIV status disclosure as a possible risk factor.”

VERSION 2 – REVIEW

REVIEWER	Simukai Shamu APHRC
REVIEW RETURNED	14-Mar-2022

GENERAL COMMENTS	The manuscript has been revised satisfactorily and meets the criteria to be accepted.